

components: income, growth, and changes in valuation multiples. If current bond yields are noticeably lower than usual, the average historical bond return will likely be of little relevance.

But not all research is based on historical averages. Sometimes researchers use different assumptions, either because they want to illustrate a particular concept with basic assumptions, or because they have incorporated their own capital market expectations.

Still, a general problem remains: safe withdrawal rates research leaves unanswered questions about what would happen if different asset classes were included, or if different assumptions were made about returns, volatilities, and correlations. My research generalized the framework for safe withdrawal rates, presenting a structure that incorporates user-specified capital market and retirement assumptions. I used Monte Carlo simulations to calculate the combinations of real portfolio arithmetic returns and volatilities that would support different withdrawal rates for various retirement durations and acceptable failure probabilities.

From there, it is a matter of determining the portfolio's expected return and volatility based on our assumptions about asset classes and their expected returns, volatilities, and correlations. In order to find the optimal asset allocation, these assumptions can be transformed into an efficient frontier. Overlaying the efficient frontier onto my exhibits lets you find the point corresponding to the highest maximum sustainable withdrawal rate. The asset allocation for the optimal point can then be determined separate from the underlying efficient frontier characteristics. This provides you with a recommended withdrawal rate and recommended asset allocation for your own specifications.

My article also described a secondary finding: that a wide range of asset allocations support withdrawal rates nearly as high as the optimal asset allocation. Many retirees will be able to support a withdrawal rate within 0.1 percentage points of the optimum with a markedly *lower* stock allocation. Conservative retirees need not be pressured into uncomfortably aggressive asset allocations, such as the recommendations for 50 to 75 percent stocks found in prominent research articles by Bill Bengen and others.